Continuity

ANSWER KEY

Testing continuity

1 Consider the piecewise function $f(x) = x^2 + 1$ for $x < 2$, and $f(x) = 3x - 1$ for $x \geq 2$.

a Find $\lim_{x \to 2^-} f(x)$. $= 5$.

b Find $\lim_{x \to 2^+} f(x)$. $= 5$.

c Is f continuous at $x = 2$? Justify. Yes — both one-sided limits equal $5 = f(2)$, so f is continuous at $x = 2$.

2 The piecewise function $f(x) = kx + 1$ for $x \leq 3$, and $f(x) = x^2 - 2$ for $x > 3$, is continuous at $x = 3$. Find the value of k.

Need $3k + 1 = 9 - 2 = 7$, so $k = 2$.

Types of discontinuity and the Intermediate Value Theorem

3 Classify the discontinuity of each function at the given point:

a $f(x) = \frac{x^2 - 1}{x - 1}$ at $x = 1$ Removable (a hole) — the factor $(x - 1)$ cancels.

b $f(x) = \frac{1}{x - 2}$ at $x = 2$ Infinite discontinuity — f blows up as $x \to 2$.

c $f(x) = \lfloor x \rfloor$ at $x = 3$ Jump discontinuity — the floor function jumps from 2 to 3.

4 Show that $f(x) = x^3 - 4x + 1$ has a root in the interval $(1, 2)$, using the Intermediate Value Theorem.

$f(1) = 1 - 4 + 1 = -2 < 0$ and $f(2) = 8 - 8 + 1 = 1 > 0$. Since f is continuous (a polynomial) and changes sign, the IVT guarantees a root in $(1, 2)$.